AI-Driven Educational Analytics System

Milestone 1 – Question Difficulty Prediction using Machine Learning & NLP

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Course: **B.Tech** CS & AI

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Mid-Sem Evaluation – Milestone 1

Problem Statement & Motivation

Problem Statement

Determining the difficulty level of exam questions is often subjective and inconsistent. Traditional methods rely heavily on instructor intuition rather than data-driven evidence.

Motivation

With increasing availability of educational data, machine learning can:

Objectively classify question difficulty

Analyze semantic complexity of questions

Integrate student performance metrics

Support intelligent assessment design

Our goal was to build an AI-driven system capable of predicting question difficulty using both textual and statistical features.

System Overview

Input

Question Text, Mean Score, Pass Rate, Standard Deviation, Discrimination Index, Question Length

Core Idea

Combine Natural Language Processing (TF-IDF) with assessment analytics to improve predictive performance.

Output

Predicted Difficulty Level: Easy, Medium, Hard

Input

The system accepts:

- Question Text
- Mean Score
- Pass Rate
- Standard Deviation
- Discrimination Index
- Question Length

Output

Predicted Difficulty Level:

- Easy
- Medium
- Hard

Dataset Description

Dataset Overview

5000

Synthetic Exam Questions

3

Difficulty Categories

Balanced across Easy, Medium, Hard

6

Features Used

Textual + Statistical

Multi-subject data including Bloom's taxonomy levels, balanced across difficulty categories.

Textual Feature

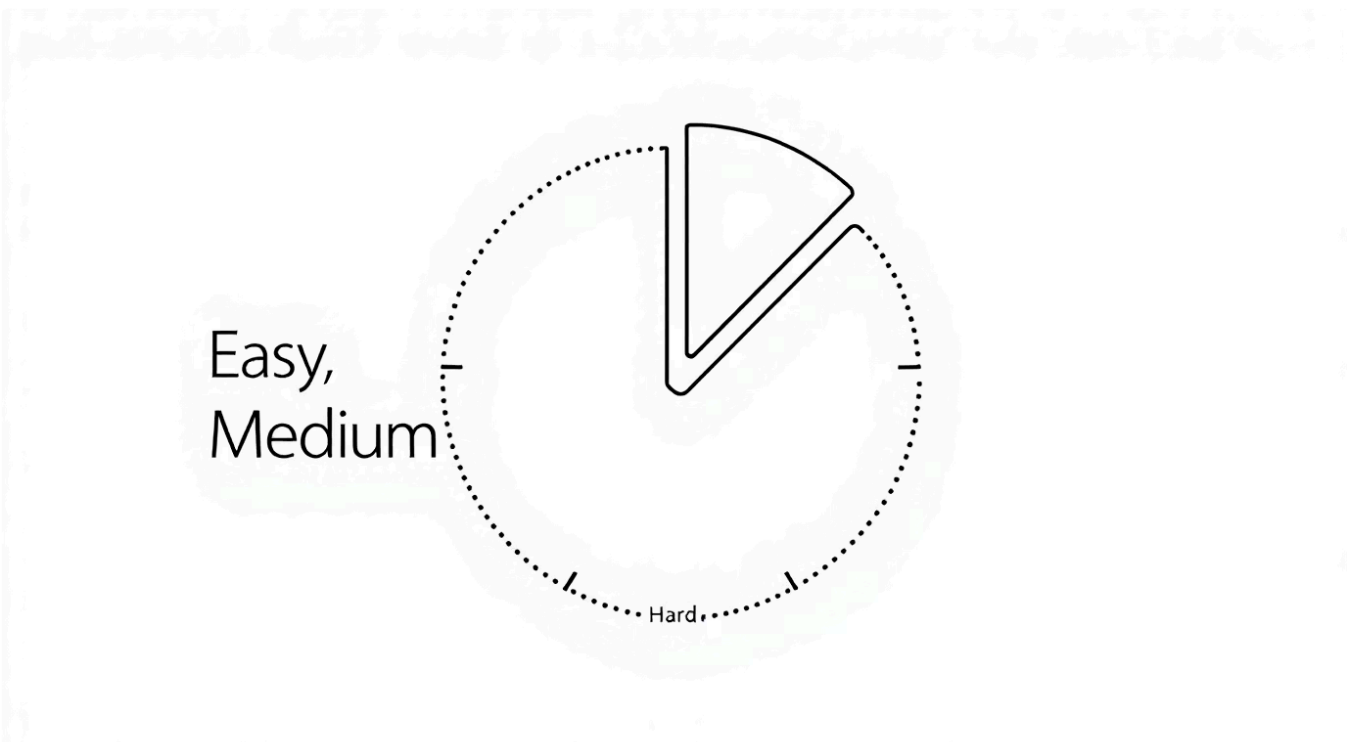
- Question text

Statistical Features

- Mean score
- Pass rate
- Standard deviation
- Discrimination index
- Question length

Target Variable

Difficulty Label (Easy / Medium / Hard)



Methodology & Preprocessing

Text Preprocessing

01

Lowercase Conversion

Conversion to lowercase

02

Punctuation Removal

Removal of punctuation

03

Noise Removal

Removal of numerical noise

04

TF-IDF Vectorization

TF-IDF assigns higher importance to informative terms and reduces the impact of common words.

Feature Engineering

We created three modeling configurations:

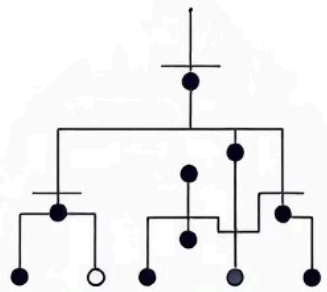
Hybrid feature stacking was performed using sparse matrix concatenation.

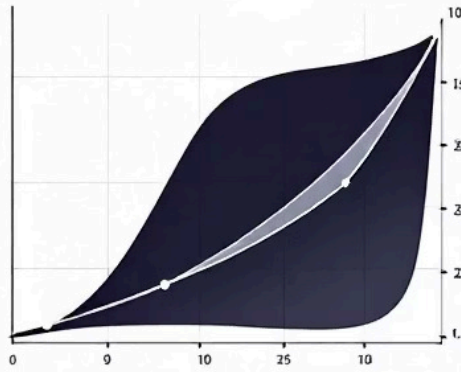
1 **Text-only features**

2 **Statistical features only**

3 **Hybrid (Text + Statistical features)**

Models Implemented





Logistic Regression

- Used for multi-class classification
- Interpretable linear model
- Tested in three configurations:

— Text-only
— Stats-only
— Combined features

Decision Tree

- Uses Gini impurity
- Captures non-linear decision boundaries
- Compared against Logistic Regression

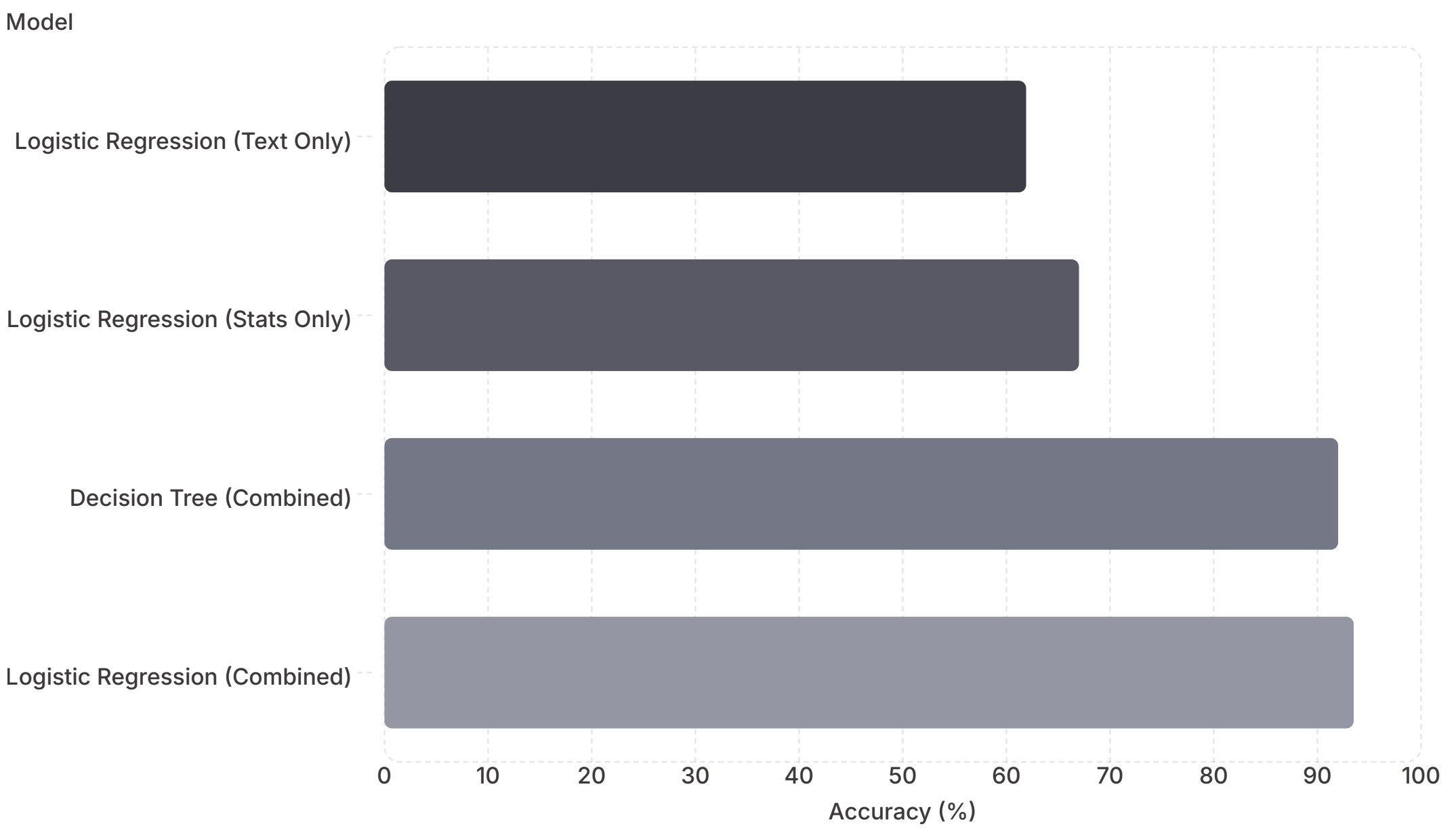
Model Evaluation & Performance

Evaluation Metrics

ACCURACY | PRECISION | RECALL | F1-SCORE | CONFUSION MATRIX

BIAS–VARIANCE ANALYSIS

Performance Comparison



Key Insight: The hybrid model significantly outperformed individual feature models, demonstrating the importance of integrating NLP with performance analytics.

Key Observations & Insights

Text-Only Limitations

Text-only model showed moderate performance (higher bias).

Statistical Boost

Statistical features improved prediction reliability.

Hybrid Strength

Hybrid model achieved strong class separation.

Hard Question Recall

Recall for "Hard" questions improved significantly in combined model.

Model Comparison

Logistic Regression slightly outperformed Decision Tree, indicating relative linear separability of feature space.

> **This validates the effectiveness of feature synergy.**

User Interface & Deployment

Deployment

Permanently hosted on **Hugging Face Spaces**

Gradio-based interface

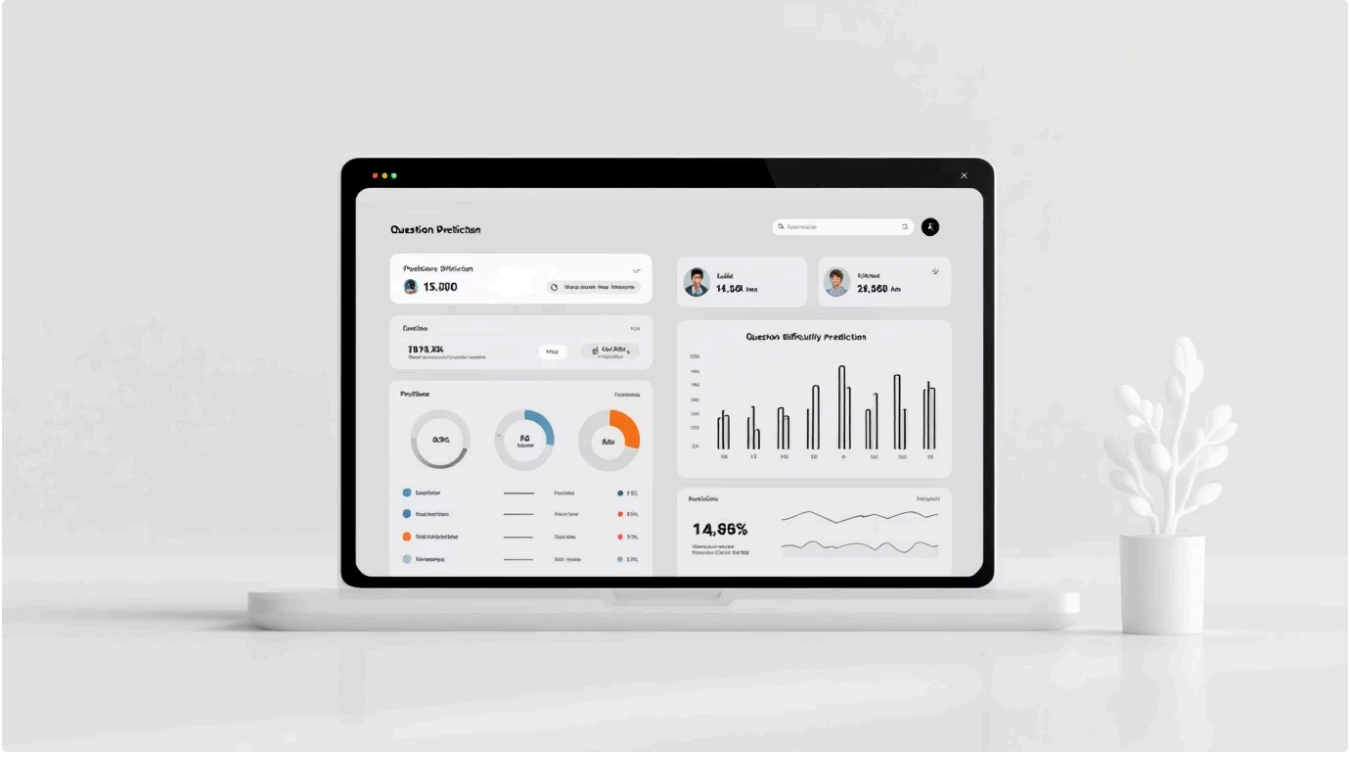
Loads **pre-trained models** for efficient inference

Interactive Interface

Two Prediction Modes:

Additional Feature:

- Flag mechanism for human-in-the-loop review.



1

Advanced Mode (High Accuracy)

Uses text + statistical features.

2

Quick Mode (Text Only)

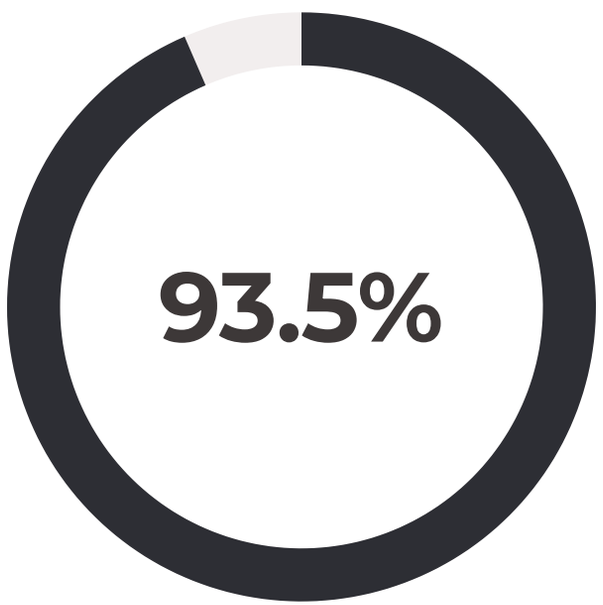
Accessible for general users without performance data.

Conclusion & Future Scope

Conclusion

This project successfully demonstrates that combining semantic NLP features with educational performance analytics significantly improves question difficulty prediction.

The hybrid Logistic Regression model achieved **93.5% accuracy**, validating the effectiveness of integrated feature modeling.



Hybrid Model Accuracy

Logistic Regression with combined features